

Report Period	Sep 26, 2026 17:40 - Sep 26, 2026 17:46
Generated At	2026-09-26 17:46:07
Classification	ML: Isolation Forest + Random Forest

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		17164	
Total Threats Mitigated	23	Total Packets Analyzed		5296257532	
High Priority	2	Malicious Packets		5285072362	
Low Priority	21	True Negatives Passed		10839	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		0	
ML Metrics	Value	SYN Flood		16	
False Positives	8	UDP Flood		2	
FP Rate	0.05%	Uncertain		5	
IF Precision	99.92%				
IF Recall	94.91%	Action		Count	
IF F1-Score	97.35%	Quarantined		4	
IF Accuracy	98.01%	Time Ban		4	
		Blackhole		0	
		Released		2	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.92%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	94.91%	Share of actual attacks successfully flagged
F1-Score	97.35%	Balanced measure combining Precision and Recall
Accuracy	98.01%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.05%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	5.09%	Actual attacks that went undetected
True Positive Rate (TPR)	94.91%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.95%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 6269	FN 336
Actual: Normal	FP 5	TN 10503

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	99.95%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	99.98%	Balanced measure combining Precision and Recall
Accuracy	99.95%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	60.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	40.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	6269	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	0

2c. Hybrid Cascaded

Metric	Value	Description
Precision	99.92%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	94.91%	Share of actual attacks successfully flagged
F1-Score	97.35%	Balanced measure combining Precision and Recall
Accuracy	98.01%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.05%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	5.09%	Actual attacks that went undetected
True Positive Rate (TPR)	94.91%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.95%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 6269	FN 336
Actual: Normal	FP 5	TN 10503

2d. Response Latency

Metric	Value	Description
Detection Time	470.8 ms	Average time taken to detect an attack
Mitigation Response Time	2.6 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.13%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	18.76%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	15.74%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.16	4	3	2026-09-26 17:42:55	2026-09-26 17:45:36
10.0.0.17	4	3	2026-09-26 17:42:58	2026-09-26 17:45:38
10.0.0.18	4	3	2026-09-26 17:42:54	2026-09-26 17:45:35
10.0.0.19	4	3	2026-09-26 17:42:58	2026-09-26 17:45:38
10.0.0.8	1	1	2026-09-26 17:43:14	2026-09-26 17:43:45
10.0.0.4	2	1	2026-09-26 17:41:58	2026-09-26 17:43:09

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-26 17:42:09	10.0.0.4	DDoS	Uncertain	59.1%	Low	Released
2026-09-26 17:42:54	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 17:42:55	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 17:42:58	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 17:42:58	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 17:43:05	10.0.0.18	DDoS	SYN Flood	92.9%	Critical	Time Ban (30s)
2026-09-26 17:43:05	10.0.0.18	DDoS	SYN Flood	92.9%	Critical	Time Ban
2026-09-26 17:43:06	10.0.0.16	DDoS	SYN Flood	91.0%	Critical	Time Ban (30s)
2026-09-26 17:43:06	10.0.0.16	DDoS	SYN Flood	91.0%	Critical	Time Ban
2026-09-26 17:43:09	10.0.0.17	DDoS	SYN Flood	93.3%	Critical	Time Ban (30s)
2026-09-26 17:43:09	10.0.0.19	DDoS	SYN Flood	92.9%	Critical	Time Ban (30s)
2026-09-26 17:43:09	10.0.0.17	DDoS	SYN Flood	93.3%	Critical	Time Ban
2026-09-26 17:43:09	10.0.0.19	DDoS	SYN Flood	92.9%	Critical	Time Ban
2026-09-26 17:43:14	10.0.0.8	DDoS	UDP Flood	85.9%	High	Time Ban (30s)
2026-09-26 17:43:45	10.0.0.8	DDoS	UDP Flood	85.9%	High	Released
2026-09-26 17:44:05	10.0.0.18	DDoS	SYN Flood	94.2%	Critical	Time Ban (1m)
2026-09-26 17:44:06	10.0.0.16	DDoS	SYN Flood	93.1%	Critical	Time Ban (1m)
2026-09-26 17:44:09	10.0.0.17	DDoS	SYN Flood	93.3%	Critical	Time Ban (1m)
2026-09-26 17:44:09	10.0.0.19	DDoS	SYN Flood	92.9%	Critical	Time Ban (1m)
2026-09-26 17:45:35	10.0.0.18	DDoS	SYN Flood	90.8%	Critical	Time Ban (2m)
2026-09-26 17:45:36	10.0.0.16	DDoS	SYN Flood	93.1%	Critical	Time Ban (2m)
2026-09-26 17:45:38	10.0.0.19	DDoS	SYN Flood	91.8%	Critical	Time Ban (2m)
2026-09-26 17:45:38	10.0.0.17	DDoS	SYN Flood	91.0%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.17	SYN Flood	0.6246	91.0%	Critical	Phase 2	1	10s	2026-09-26 17:45:38	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6552	91.8%	Critical	Phase 2	1	10s	2026-09-26 17:45:38	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6345	93.1%	Critical	Phase 2	1	11s	2026-09-26 17:45:36	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6462	90.8%	Critical	Phase 2	1	11s	2026-09-26 17:45:35	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.19	SYN Flood	0.6466	92.9%	Critical	Phase 2	3	1m 11s	2026-09-26 17:44:09	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6409	93.3%	Critical	Phase 2	3	1m 11s	2026-09-26 17:44:09	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6310	93.1%	Critical	Phase 2	3	1m 11s	2026-09-26 17:44:06	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6270	94.2%	Critical	Phase 2	3	1m 11s	2026-09-26 17:44:05	Attack persisted - escalating to time ban
10.0.0.8	UDP Flood	0.5719	85.9%	High	Phase 2	1	31s	2026-09-26 17:43:45	Ban Expired
10.0.0.19	SYN Flood	0.6142	92.9%	Critical	Phase 2	2	41s	2026-09-26 17:43:39	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6020	93.3%	High	Phase 2	2	41s	2026-09-26 17:43:39	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6027	93.1%	High	Phase 2	2	41s	2026-09-26 17:43:36	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6425	94.2%	Critical	Phase 2	2	41s	2026-09-26 17:43:35	Attack persisted - escalating to time ban
10.0.0.4	Uncertain	0.5859	61.4%	Low	Phase 1	1	11s	2026-09-26 17:43:09	Attack Stopped
10.0.0.19	SYN Flood	0.6516	92.9%	Critical	Phase 2	1	11s	2026-09-26 17:43:09	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6398	93.3%	Critical	Phase 2	1	11s	2026-09-26 17:43:09	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6389	91.0%	Critical	Phase 2	1	11s	2026-09-26 17:43:06	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6464	92.9%	Critical	Phase 2	1	11s	2026-09-26 17:43:05	Attack persisted - escalating to time ban
10.0.0.4	Uncertain	0.6239	59.1%	Low	Phase 1	1	11s	2026-09-26 17:42:09	Attack Stopped

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE